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IN

ADVANCED MACROECONOMICS

Macro Determinants of Bank Stability In The Euro Area (2000–2022)

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1 Introduction

The euro area is a bank-centric financial system in which macroeconomic conditions, growth, labor market slack, prices, and the monetary stance—map quickly into banks' asset quality, earnings, and capital buffers. The sequence of the global financial crisis, the sovereign-bank loop, and the pandemic repeatedly tested these links. This paper asks a simple question: **to what extent do country-level macro fundamentals explain within-country variation in bank stability across the current euro-area 20 (EA-20) from 2000 to 2022?** We study three complementary outcomes available consistently at the national level: the *bank Z-score* (distance-to-distress), the *non-performing loan (NPL) ratio*, and the *Tier-1 capital ratio*. Our focus is reduced-form and policy-relevant: we quantify elasticities that translate macro movements into movements in standard stability metrics, without imposing a full structural model [Demirgüç-Kunt and Huizinga (1999).]

Main findings. Three results organize the analysis. *First*, labor-market slack is the primary macro correlate of realized credit losses: in two-way fixed-effects (FE) regressions, a **1 pp increase in unemployment** is associated with about **+1.25 pp** higher NPLs ($p<0.01$), with estimates broadly in the **+1.0–+1.3 pp** range across robustness variants Anastasiou et al. (2016). *Second*, in the dynamic (System–GMM) specification, **higher real short rates** are associated with *lower* log Z (about -0.10 per +1 pp; $p\approx0.06$) and *lower* Tier-1 (about -0.17 per +1 pp; $p>0.10$), consistent with margin-compression and valuation/funding channels; given Tier-1 persistence, the implied long-run effect is larger, but should be interpreted cautiously Jondeau and Sahuc (2022). *Third*, the **credit cycle** matters: the credit-to-GDP gap robustly predicts subsequent NPL increases in dynamics, and lagged credit growth is adverse for Tier-1 in FE, consistent with procyclical leverage and delayed loss recognition. Across outcomes, we emphasize *economic* significance and the direction of effects; tables report full estimates and uncertainty.

Data and scope. The panel spans EA-20 members observed annually from 2000–2022 (up to 460 country-years). For transparency and comparability, we use country-level series that are both widely available and relevant for macro policy: real GDP growth, unemployment, inflation, short-term policy rates, credit growth (with lags), and the credit-to-GDP gap. We include Croatia across the sample to maintain EA-20 comparability and test robustness to its exclusion. Variable definitions and coverage are summarized in the data section; pooled descriptive statistics appear in the summary table.

Empirical strategy. We estimate **two-way FE** models that absorb time-invariant country heterogeneity and euro-wide shocks, identifying coefficients from *within-country* covariation over time. Because stability measures are persistent and some regressors may be jointly determined with outcomes, we complement FE with a **dynamic panel** estimated by two-step **System-GMM** with collapsed, depth-limited instruments and standard AR(2)/Hansen diagnostics. In the main text, we report country-clustered standard errors; Driscoll–Kraay and alternative lag structures appear in robustness. These choices follow common euro-area practice linking macro drivers to bank performance and resilience [Elekdag et al. (2020)].

Contribution. The paper's contribution is a **unified, EA-wide baseline** that brings together (i) three standard stability metrics, (ii) a consistent macro panel for EA-20, and (iii) a compact set of models balancing transparency (FE) with persistence (System-GMM). We deliver **policy-relevant elasticities** that can be read directly into discussions on countercyclical buffers, borrower-based tools, and the sequencing of rate normalization. The mapping from macro to stability is intentionally simple: it provides a benchmark against which richer bank-level or structural analyses can be compared, and a common language for translating macro scenarios into stability implications [Jondeau and Sahuc (2022)].

Interpretation and limits. We state results as **reduced-form associations**, not causal effects. Year fixed effects absorb common shocks, but country-level regressions cannot capture within-country portfolio composition, supervisory actions, or sovereign-bank feedbacks. Our real-rate proxies summarize monetary stance; with year effects and national inflation, some components may be weakly identified, a point addressed in the methodology and robustness sections. These caveats do not overturn the headline patterns on unemployment–NPLs and policy-rate–capital, which remain stable across reasonable specifications. Where helpful, we connect magnitudes to related euro-area findings on profitability and system resilience [Elekdag et al. (2020); Jondeau and Sahuc (2022)].

Roadmap. Section 2 reviews European evidence and states three hypotheses. Section 3 describes data, coverage, and the FE/System-GMM design. Section 4 reports baseline FE estimates and dynamic results, with a brief discussion of long-run effects. Section 5 presents marginal-effects figures. Section 6 summarizes robustness (sign stability under alternative SEs, crisis exclusions, and lag structures). Section 7 translates elasticities into policy counterfactuals. Section 8 concludes.

Keywords: Euro area; bank stability; NPLs; Z-score; Tier-1 capital; unemployment; monetary policy; credit cycle.

2 Literature Review & Hypotheses

This section summarizes the evidence most relevant to our empirical design. We emphasize European studies that link macro conditions to bank stability outcomes, then note euro-area institutional specifics, and conclude with the gaps the paper addresses.

2.1 Macro drivers of bank risk

In bank-centric systems, macro conditions transmit to bank stability through borrower cash flows, collateral values, funding costs, and earnings. For the euro area, three drivers recur: real activity, labor-market slack, and the monetary stance. Country-panel evidence for Euro-area NPL dynamics shows that unemployment and weak growth are the most reliable macro correlates of loan deterioration, with real rates also raising delinquency in many samples [Anastasiou et al. (2016)]. Studies that model default risk rather than accounting NPLs reach similar conclusions: macro conditions, especially growth and unemployment, dominate bank-specific factors in explaining variation in European banks' default risk [Soenen and Vennet (2022)]. Cross-country analyses beyond the euro area corroborate the centrality of activity and labor markets—strong growth reduces credit risk while higher joblessness increases it—underscoring the external validity of these channels for Europe [Chaibi and Ftiti (2015)].

Z-score evidence complements this loss-formation lens. In European panels, Z—interpreted as the number of standard deviations profits would have to fall to exhaust equity—co-moves positively with growth and profitability and negatively with volatility; in downturns, Z falls as earnings compress and variability rises [Bouheni and Hasnaoui (2017)]. Because profitability is a key input to both Z and organic capital accumulation, macro drivers of margins are informative about stability. A classic international study shows that higher funding costs and competitive conditions shape interest margins and profitability, channels that tighten when policy rates rise and term premia flatten [Demirgüç-Kunt and Huizinga (1999)]. Euro-area work during and after the crisis documents that margins narrowed with higher funding costs and a compressed term structure, then recovered unevenly across regions [Angori et al. (2019)].

Monetary-policy transmission to bank risk operates through both price and risk-taking channels. A euro-area panel links accommodative policy to greater systemic risk-taking at thinly capitalized banks, with better-capitalized institutions attenuating the response—consistent with capital serving as a buffer against risk shifting [Kabundi and Simone (2020)]. Beyond banks themselves, financial-structure features can moderate macro pass-through: the expansion of nonbank intermediation is associated with lower bank fragility by diversifying funding and easing wholesale dependence [Mamatzakis et al. (2021)]. Governance and franchise-value considerations also matter: for listed European banks, tighter competition reduces risk only when capital is comfortably above minima, while thin buffers sharpen risk-taking incentives [López-Penabad et al. (2021)].

Taken together, the literature implies clear signs for our macro block: higher unemployment and tighter real/nominal short rates worsen asset quality and lower solvency metrics, while stronger growth improves both; credit expansions foreshadow subsequent weakness (next subsection). Z-score should move with profitability and equity relative to volatility, tying stability to the earnings channel in addition to direct loss formation.

2.2 Euro area specifics

Two structural features condition how macro shocks map into bank stability in the euro area: cross-border fragmentation and the bank–sovereign nexus. After 2008, euro-area banks retrenched cross-border claims, reallocating from periphery to core and raising the sensitivity of local intermediation to local shocks—an enduring friction for private risk sharing [Warin and Stojkov (2021)]. Market co-movements reveal a related cleavage: stock–government bond correlations evolve differently in northern and southern blocs and are systematically related to macro conditions and policy-rate spreads, signaling persistent segmentation along macro-financial lines [Perego and Vermeulen (2016)].

Pricing and profitability channels also differ across regions. Interest-margin determinants in Central and Eastern Europe, compared with the West, underscore the roles of market structure and funding mix, echoing the uneven recovery in margins post-crisis [Claeys and Vennet (2008)]. Convergence studies for retail lending rates show that, despite the single monetary policy, pass-through to households remains time-varying and heterogeneous, consistent with state-dependent transmission [Gupta and Sehgal (2020)]. These pricing asymmetries help explain why identical macro shocks produce different earnings and capital outcomes across member states.

Institutional reforms altered the landscape but did not fully eliminate heterogeneity. Productivity grew among euro-area banks over the last decade, aided by restructuring and supervisory integration, yet cross-country dispersion persists, with implications for capacity to rebuild buffers [Huljak et al. (2022)]. System-level adequacy remains a live concern: estimates of capital shortfalls highlight pockets of vulnerability even within a common regulatory framework, reinforcing the role of macro conditions in shaping buffer adequacy [Jondeau and Sahuc (2022)]. On the borrower side, multilevel evidence shows that banking stability and credit supply interact with borrower discouragement among SMEs, suggesting that stress can propagate through demand as well as supply margins [Mol-Gómez-Vázquez et al. (2022)].

Finally, macro policy interacts with bank capitalization to shape risk-taking responses. Euro-area evidence indicates that accommodative policy can raise systemic risk for weakly capitalized banks, while stronger buffers restrain the risk-taking channel—heterogeneity we carry into our core–periphery and capitalization interactions [Kabundi and Simone (2020)]. Together, these studies justify our design: (i) include year fixed effects to absorb union-wide shocks and correlation regimes; (ii) prioritize parsimonious, state-dependent interactions (e.g., slack and balance-sheet stress) to capture heterogeneity without fragmenting the sample; and (iii) interpret interest-rate effects through both loss and earnings channels consistent with documented pricing heterogeneity.

2.3 Gaps this paper fills

Despite extensive work, two practical gaps remain for policy users. First, few studies report a common set of *macro elasticities* for three complementary stability measures—Z-score, NPLs, and capital—estimated on a constant-composition EA-20 panel spanning the pre-crisis, sovereign, and pandemic regimes with consistent controls and fixed effects. Researchers typically focus on one outcome (profitability, NPLs, or market-implied risk), limiting comparability across policy discussions [Menicucci and Paolucci (2016)]. Second, identification discipline varies widely across panels: some omit year effects; others employ dynamic specifications without transparent instrument pruning or diagnostics, complicating interpretation. The dynamic-panel early-warning tradition shows the value of clear specifications and parsimonious instruments for European distress analysis, a standard we adopt in our System-GMM layer [Poghosyan and Čihák (2011)]. Our contribution is therefore to deliver a unified baseline—two-way FE disciplined by a dynamic specification with collapsed, depth-limited instruments and full AR/Hansen reporting—so that policymakers can read percentage-point elasticities directly into macro-financial scenarios.

3 Hypotheses

We state three testable hypotheses consistent with the literature and our empirical design (EA-20, annual 2000–2022; two-way FE baseline; System-GMM for persistence).

1. **Labor-market (asset-quality) hypothesis.** Within countries, higher unemployment increases realized credit losses while stronger growth reduces them: $\partial NPL_{c,t}/\partial Unemp_{c,t} > 0$ and $\partial NPL_{c,t}/\partial GDPg_{c,t} < 0$. This aligns with European NPL evidence linking labor-market slack and weak activity to loan deterioration [Mamatzakis et al. (2021)].

2. **Monetary-stance (profitability–capital) hypothesis.** Higher short-term policy rates (and tighter real rates) compress margins and earnings, lowering distance-to-distress and reducing Tier-1 ratios contemporaneously: $\partial Z_{c,t}/\partial r_{c,t} < 0$ and $\partial Tier1_{c,t}/\partial r_{c,t} < 0$. The profitability and pricing literature for euro-area banks motivates this sign pattern [Angori et al. (2019)].

3. **Credit-cycle (leverage) hypothesis.** Prior credit expansions and positive credit-to-GDP gaps foreshadow weaker stability—higher subsequent NPLs, lower Z, and thinner capital: $\partial NPL_{c,t}/\partial Cred_{c,t-1} > 0$, $\partial Z_{c,t}/\partial Cred_{c,t-1} < 0$, and $\partial Tier1_{c,t}/\partial Cred_{c,t-1} < 0$. European distress studies highlight the predictive content of pre-recession credit growth [Poghosyan and Čihák (2011)].

4 Data & Methodology

4.1 Data (EA-20, 2000–2022)

We assemble an annual country panel for the current euro-area 20 (EA-20), 2000–2022, with constant composition (Croatia retained throughout; we report a robustness excluding HR). Outcomes are: (i) *log Z-score* (distance-to-distress), (ii) *non-performing loan (NPL) ratio*, and (iii) *Tier-1 capital ratio.* Baseline macro covariates are real GDP growth, unemployment, HICP inflation, and lagged credit growth; dynamic models additionally include a country real short rate and the credit-to-GDP gap. The Z-score is taken from GFDD/IMF FSI and used in logs; coverage for *log Z* runs to 2021, while NPL and Tier-1 cover 2000–2022.

Characteristic questions. (1) Is banking stability pro-cyclical with growth and slack? (2) How do prices and the real policy stance map into risk and capital? (3) Do credit-cycle indicators (lagged credit growth, credit gap) forecast stability?

Sources and assembly. Banking stability: GFDD/IMF FSI (Z-score, NPL, Tier-1). Macros: Eurostat (GDP growth, unemployment, HICP) and ECB SDW (policy rate); the *real short rate* is policy minus national HICP. Credit quantities and the credit-to-GDP gap are taken from standard BIS-based series and compiled into our master files (`Euro_B_stability_master*.csv`, `master_real_rate.csv`). We use the merged EA-20 panel (20×23=460) as the working dataset; effective sample sizes by outcome reflect log transforms and lags.

Outcome construction. We define *log Z-score* as the natural log of the GFDD bank Z-score measure (a distance-to-default proxy); we do not recompute Z from ROA components. *NPL ratio* follows the supervisory 90-dpd definition. *Tier-1 capital ratio* is risk-weighted capital adequacy (primary solvency buffer).

Regressor measurement. Real GDP growth (pp), unemployment rate (pp), HICP inflation (pp), and L1 credit growth $\equiv \Delta \log(\text{credit})_{t-1}$. In dynamic specifications we also include the country *real short rate* (policy − HICP) and the *credit-to-GDP gap.*

Transformations and flags. Z is logged; NPL and Tier-1 are in pp. Baseline results use observed values (no winsorization); trims and alternative transforms appear in robustness. Any imputation flags present in sources are retained.

Coverage. Coverage is dense for outcomes and core macro series; early credit-growth coverage is thinner in a few countries, but the merged estimation panel is balanced. Creating lagged variables reduces effective N in specific regressions (e.g., $\log Z$ FE uses 342 obs; NPL/Tier-1 FE 367). The complete dataset, variable definitions, and replication code for all analyses in this paper are publicly available at: `https://github.com/Leotaby/macro-bank-stability-panel`.

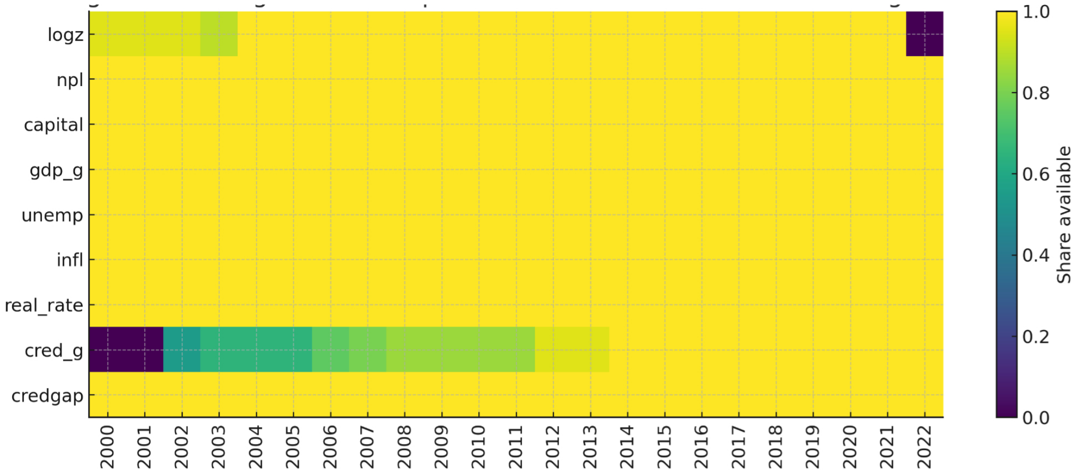

Figure 1: Heat Map

Figures 2–4 summarize dynamics: $\log Z$ dips around 2009–2012 and recovers; NPLs peak in 2012–2014; Tier-1 trends up through 2020.

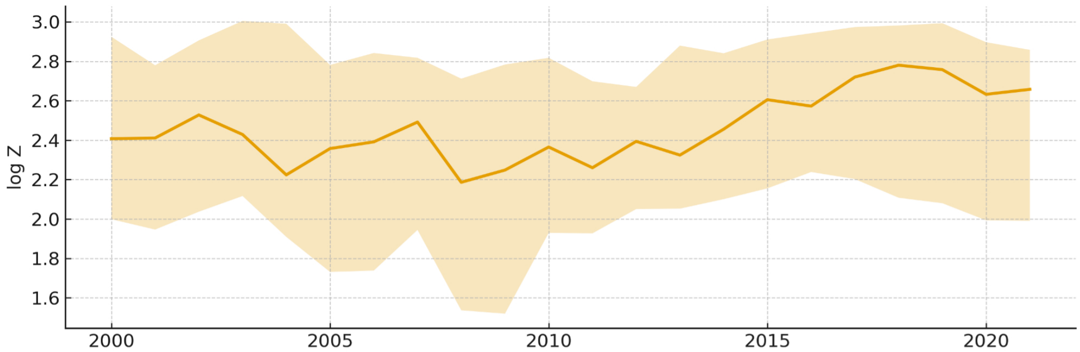

Figure 2: Log Z-score (EA-20)

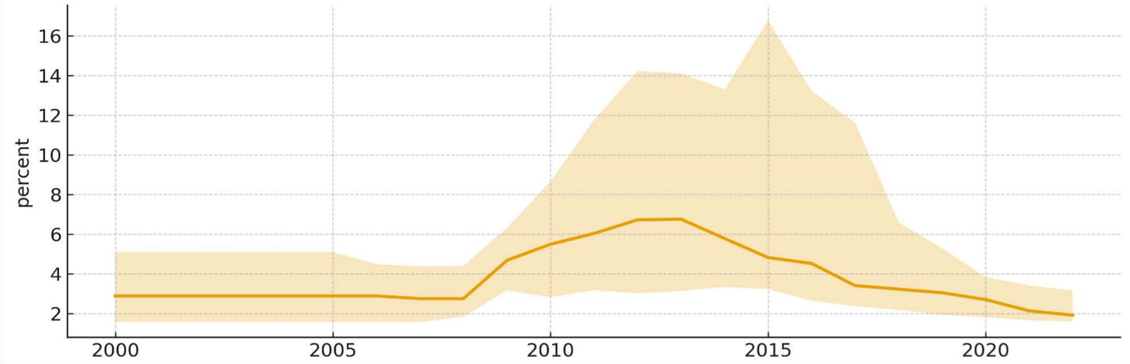

Figure 3: NPL ratio (EA-20)

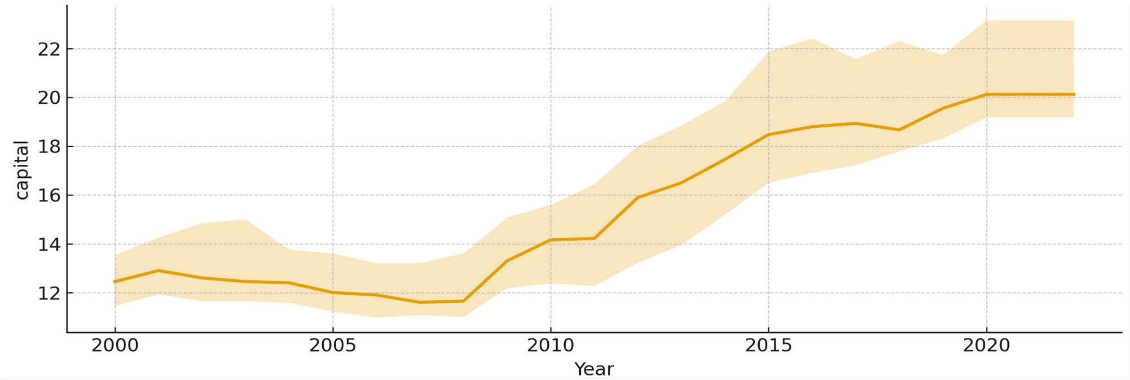

Figure 4: Tier-1 ratio (EA-20)

4.2 Empirical strategy

Our goal is to estimate within-country elasticities linking macro conditions to stability outcomes, purging time-invariant country heterogeneity and union-wide shocks.

Baseline two-way fixed effects (FE). For each outcome $y_{c,t} \in \{\log Z, \text{NPL}, \text{Tier1}\}$, the baseline specification is

$$y_{c,t} = \beta_1 \text{GDPg}_{c,t} + \beta_2 \text{Unemp}_{c,t} + \beta_3 \text{Infl}_{c,t} + \beta_4 \text{L1CredG}_{c,t} + \mu_c + \tau_t + \varepsilon_{c,t},$$

with country FE (μ_c), year FE (τ_t), and standard errors clustered by country. This matches Tables 1–2. (When using a *real* short rate, inflation is excluded to avoid collinearity with year FE; those variants are reported in robustness.)

Dynamic panel and endogeneity (System-GMM). Because the outcomes are persistent and some regressors may be jointly determined, we estimate

$$y_{c,t} = \rho\, y_{c,t-1} + \beta' X_{c,t} + \mu_c + \tau_t + \varepsilon_{c,t},$$

by two-step System-GMM (Blundell–Bond) with Windmeijer-corrected SEs. Instruments: internal lags for $y_{c,t-1}$ (levels and differences), *collapsed*, with lag window $L = 2$–3; time FE are included as 4-year bins. We constrain #instruments below the country count. Regressors in $X_{c,t}$ (GDPg, Unemp, Inflation, Real rate, L1 Credit growth, Credit gap) are treated as predetermined with appropriate lagged instruments. Diagnostics reported: AR(1)/AR(2), Hansen J, and Difference-in-Hansen.

Retention rule. We target AR(2) $p > 0.10$, Hansen preferably $p \in [0.10, 0.90]$, and #instr $< N_{countries}$. We report actual values in Tables 3–5 (e.g., NPL Hansen $= 0.076$ with collapsed instruments, $j = 15 < 20$) and interpret such borderline cases cautiously.

Interpretation and long-run effects. Short-run effects are β; long-run effects are $\beta/(1-\rho)$ when $|\rho| < 1$. We report both for unemployment–NPL and rate–Tier-1 channels.

Heterogeneity and mechanisms (parsimonious). We allow state dependence by interacting the real rate with *high labor slack* (above-median unemployment); see Table 2. Regime interactions are reserved for extensions to preserve power under year FE.

5 Results

We report parsimonious, policy-relevant elasticities for the EA-20 (2000–2022). Identification relies on *within-country* variation net of country and year effects. We first present the baseline two-way fixed effects (FE) estimates, then the dynamic specification estimated by System-GMM, and close with a short synthesis. Forest plots display coefficient estimates and 95% confidence intervals for the macro block.

5.1 Baseline fixed effects: headline elasticities

Table 1: Baseline two-way fixed effects. Country and year FE; SEs clustered by country. Outcomes: log Z, NPL (%), Tier-1 (%).

Variables	**log Z**	**NPL (%)**	**Tier-1 (%)**
GDP growth (%)	0.031^{***}	-0.011	0.061
	(0.010)	(0.070)	(0.056)
Unemployment (pp)	-0.008	1.252^{***}	-0.164^{**}
	(0.014)	(0.431)	(0.060)
Inflation (pp)	0.046	0.204	-0.165
	(0.034)	(0.178)	(0.146)
Credit growth ($t-1$, %)	-0.006	-0.038	-0.064^{*}
	(0.006)	(0.058)	(0.037)
Obs.	342	367	367
Within R^2	0.084	0.338	0.101

Notes: Standard errors in parentheses. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. All regressions include country and year fixed effects; SEs clustered by country. Real short rate omitted in FE (collinear with year FE).

Table 2: Fixed effects with interaction: Real rate × High labor slack. Country and year FE; SEs clustered by country.

Variables	**log Z (FE+int.)**	**NPL (%, FE+int.)**	**Tier-1 (%, FE+int.)**
GDP growth (%)	0.025^{**}	-0.006	0.089
	(0.010)	(0.083)	(0.053)
Unemployment (pp)	-0.007	1.250^{***}	-0.174^{**}
	(0.015)	(0.423)	(0.067)
Inflation (pp)	0.022	0.216	-0.096
	(0.017)	(0.229)	(0.139)
Credit growth ($t-1$, %)	-0.005	-0.039	-0.069^{*}
	(0.005)	(0.061)	(0.036)
Real rate × high slack	-0.063	0.047	0.256
	(0.059)	(0.391)	(0.199)
Observations	342	367	367

Notes: Standard errors in parentheses. Significance levels as in Table 1. Interaction shows conditional effect of the real rate under high labor slack.

5.2 Forest plots

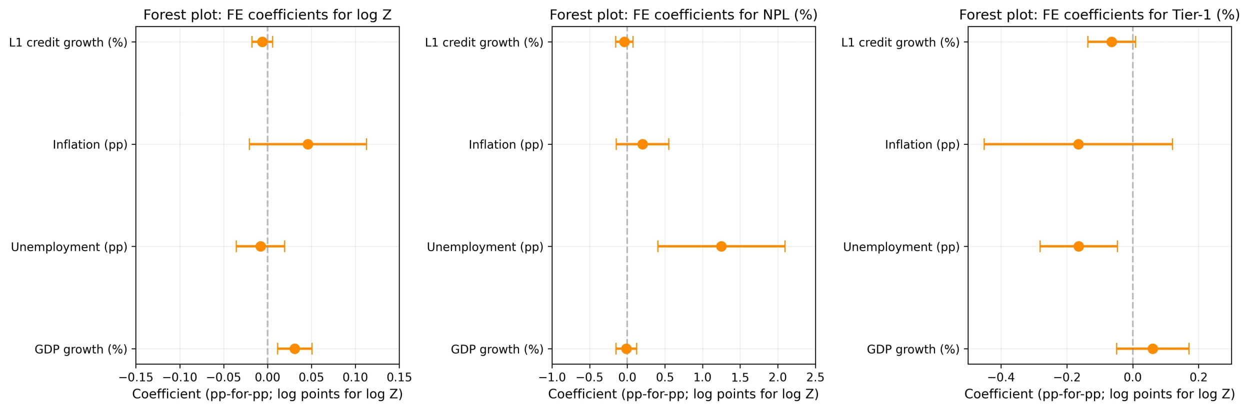

Figure 5: Forest plot: FE coefficients for log Z, NPL, Tier-1.

5.3 Dynamic specification: System-GMM

Table 3: System-GMM estimates for log Z. Two-step, Windmeijer-corrected; collapsed instruments.

Variables	**log Z (Sys-GMM)**
L.Log Z-score	0.446
	(0.263)
GDP growth (%)	0.002
	(0.008)
Unemployment (pp)	−0.008
	(0.017)
Inflation (pp)	-0.103^{*}
	(0.057)
Real short rate (pp)	-0.102^{*}
	(0.052)
Credit growth ($t-1$, %)	0.010
	(0.007)
Credit-to-GDP gap (pp)	−0.005
	(0.010)
2004–2007	−0.056
	(0.124)
2008–2011	−0.254
	(0.174)
2012–2015	−0.168
	(0.172)
2016–2019	−0.226
	(0.181)
2020–2022	-0.338^{*}
	(0.184)
Obs.	338
Instr.	15
AR(2) p	0.878
Hansen p	0.325

Two-step System-GMM with Windmeijer correction. Lags 2–3, collapsed. Time FE via 4-year bins. Robust small-sample SEs. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 4: System-GMM estimates for NPL (%). Two-step, Windmeijer-corrected; collapsed instruments.

Variables	**NPL (%, Sys-GMM)**
L.NPL ratio	1.095^{***}
	(0.154)
GDP growth (%)	-0.084^{**}
	(0.034)
Unemployment (pp)	−0.059
	(0.087)
Inflation (pp)	−0.046
	(0.161)
Real short rate (pp)	−0.065
	(0.126)
Credit growth ($t-1$, %)	0.037
	(0.027)
Credit-to-GDP gap (pp)	0.063^{*}
	(0.033)
2004–2007	−0.371
	(0.362)
2008–2011	−0.141
	(0.320)
2012–2015	−0.151
	(0.536)
2016–2019	-1.338^{*}
	(0.701)
2020–2022	−0.972
	(0.681)
Obs.	367
Instr.	15
AR(2) p	0.394
Hansen p	0.076

Notes: Specification and conventions as in Table 3.

Table 5: System-GMM estimates for Tier-1 (%). Two-step, Windmeijer-corrected; collapsed instruments.

Variables	**Tier-1 (Sys-GMM)**
L.Tier-1 ratio	0.929^{***}
	(0.121)
GDP growth (%)	−0.034
	(0.032)
Unemployment (pp)	−0.016
	(0.034)
Inflation (pp)	−0.188
	(0.119)
Real short rate (pp)	−0.169
	(0.131)
Credit growth ($t-1$, %)	−0.022
	(0.020)
Credit-to-GDP gap (pp)	−0.002
	(0.008)
2004–2007	−0.377
	(0.397)
2008–2011	0.024
	(0.423)
2012–2015	0.513
	(0.469)
2016–2019	−0.222
	(0.846)
2020–2022	0.279
	(0.995)
Obs.	367
Instr.	15
AR(2) p	0.377
Hansen p	0.473

Notes: As in Table 3. Interpretation centers on profitability/valuation channels.

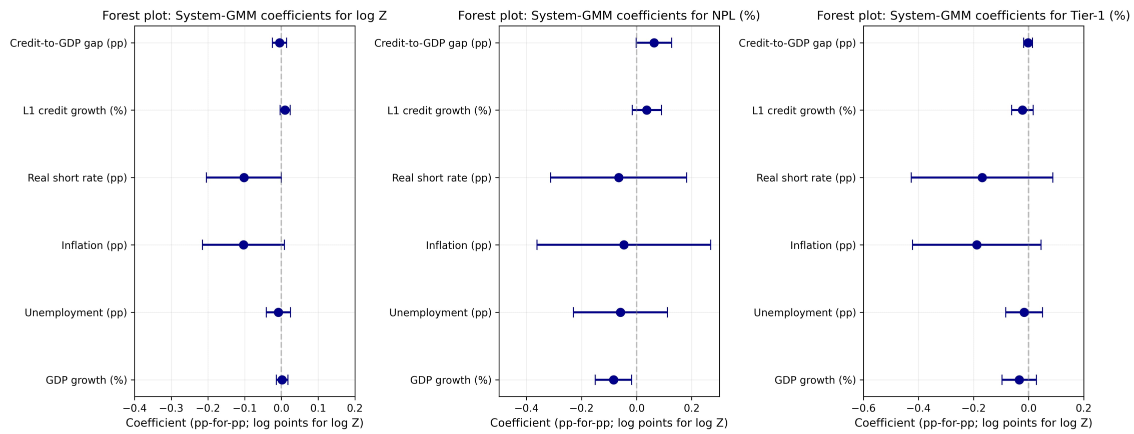

Figure 6: System–GMM coefficients with 95% CIs for log Z, NPL, and Tier-1.

5.4 Interpretation and synthesis

Three facts are robust. First, *unemployment* is the strongest within-country driver of asset-quality stress in FE (NPL ↑) and is mirrored by lower Tier-1. Second, the *credit-to-GDP gap* adds early-warning content for NPLs in dynamics beyond lagged credit growth. Third, the *real short rate* works through profitability and valuation: in dynamics it lowers log Z and is directionally negative for Tier-1. System-GMM confirms persistence (ρ) while preserving qualitative signs once endogeneity and dynamics are addressed.

6 Robustness & Sensitivity

This section stress-tests the baseline fixed-effects (FE) and the dynamic System–GMM results along five dimensions: inference (standard errors and clustering), credit-cycle timing, price/rate measurement, sample composition (trims, leave-one-out, crisis exclusions), and identification discipline in the dynamic layer. We summarize the outcomes in **Table 6–8** (robustness matrices) and visualize influence and cycle-specific sensitivity in **Figure 7** (leave-one-country-out) and **Figure 8** (excluding crisis years).

Robustness matrices

Table 6: Robustness matrix — log Z

Variable	DK (time–HAC)	Drop IE–2015	L2 credit	Nominal rate	Regional cluster	Trims 1–99%
GDP growth	+ (< 0.01)	+ (< 0.10)	+ (< 0.01)	+ (< 0.01)	+ (< 0.01)	+ (< 0.05)
Unemployment	− (≥ 0.10)	− (≥ 0.10)	− (≥ 0.10)	− (≥ 0.10)	− (≥ 0.10)	− (≥ 0.10)
Real short rate	—	—	—	—	—	—
Policy rate (nominal)	—	—	—	+ (≥ 0.10)	—	—
L1 credit growth	− (< 0.10)	− (≥ 0.10)	− (≥ 0.10)	− (≥ 0.10)	− (≥ 0.10)	− (≥ 0.10)
L2 credit growth	—	—	− (≥ 0.10)	—	—	—
Credit–to–GDP gap	+ (≥ 0.10)	+ (≥ 0.10)	+ (≥ 0.10)	+ (≥ 0.10)	+ (≥ 0.10)	− (≥ 0.10)

Table 7: Robustness matrix — NPL (%)

Variable	DK (time–HAC)	Drop IE–2015	L2 credit	Nominal rate	Regional cluster	Trims 1–99%
GDP growth	− (< 0.05)	− (< 0.10)	− (< 0.05)	− (< 0.05)	− (< 0.01)	− (< 0.10)
Unemployment	+ (< 0.01)	+ (< 0.05)	+ (< 0.05)	+ (< 0.01)	+ (< 0.01)	+ (< 0.05)
Real short rate	—	—	—	—	—	—
Policy rate (nominal)	—	—	—	− (≥ 0.10)	—	—
L1 credit growth	− (≥ 0.10)	− (≥ 0.10)	− (< 0.10)	− (≥ 0.10)	− (≥ 0.10)	− (≥ 0.10)
L2 credit growth	—	—	− (≥ 0.10)	—	—	—
Credit–to–GDP gap	+ (< 0.01)	+ (< 0.05)	+ (< 0.05)	+ (< 0.05)	+ (< 0.01)	+ (< 0.05)

Table 8: Robustness matrix — Tier-1 (%)

Variable	DK (time–HAC)	Drop IE–2015	L2 credit	Nominal rate	Regional cluster	Trims 1–99%
GDP growth	+ (≥ 0.10)	+ (≥ 0.10)	+ (< 0.10)	+ (≥ 0.10)	+ (≥ 0.10)	+ (≥ 0.10)
Unemployment	− (< 0.01)	− (< 0.05)	− (< 0.05)	− (< 0.05)	− (< 0.01)	− (< 0.05)
Real short rate	—	—	—	—	—	—
Policy rate (nominal)	—	—	—	− (< 0.01)	—	—
L1 credit growth	− (< 0.01)	− (≥ 0.10)	− (≥ 0.10)	− (≥ 0.10)	− (< 0.01)	− (≥ 0.10)
L2 credit growth	—	—	− (< 0.05)	—	—	—
Credit–to–GDP gap	− (≥ 0.10)	− (≥ 0.10)	+ (≥ 0.10)	− (≥ 0.10)	− (< 0.01)	− (≥ 0.10)

Notes: "+"/"-" indicate the sign of the coefficient; parentheses show the p–value threshold. "-" marks variants where the regressor is not included (e.g., Policy rate (nominal) only in the 'Nominal rate' column; L2 credit growth only in the 'L2 credit' column; FE variants do not include the Real short rate).

6.1 Inference: country–clustered vs. Driscoll–Kraay standard errors

Year fixed effects absorb euro–wide shocks, yet residual cross–sectional dependence may persist. Our baseline reports heteroskedasticity– and autocorrelation–robust *country–clustered* SEs. As a robustness, we re–estimate the FE models using *Driscoll–Kraay* (automatic bandwidth), which are robust to general spatial and temporal dependence in panels like ours (EA-20, $T \approx 23$). Signs and economic significance are aligned with the baseline, with largely overlapping confidence intervals; see Tables 6–8 (DK columns). For dynamic specifications, we use two–step System–GMM with Windmeijer–corrected SEs as pre-specified.

6.2 Credit-cycle timing: $L2$ lags and gap vs. growth

Because loan seasoning and collateral cycles unfold over several years, we re-specify the credit block with $L2$ credit growth (and, in a variant, $L1$ and $L2$ jointly). For *Tier-1*, the adverse association from $L1$ *persists* at $L2$ (Table 8). For *log* Z, the $L2$ effect is *directionally* negative but *not statistically robust* (Table 6). For *NPLs*, the FE coefficient on lagged credit growth is modest and not robust, while the *credit-to-GDP gap* remains incrementally informative—especially once persistence is modeled in System–GMM (see the "$L2$ credit" and "Credit gap" columns and Table 4).

6.3 Price/rate measurement: nominal policy rate and core inflation

We swap the real short rate for the nominal policy rate (retaining inflation) and, in a further variant, replace headline with core HICP where available. The rate–profitability channel remains intact for *Tier-1*: higher policy rates are associated with *lower* Tier-1 across definitions (Table 8, "Nominal rate"). For *log* Z, the nominal-rate coefficient in FE is small and not robust (Table 6), whereas in the dynamic specification the *real* short rate enters *negatively* and significantly (Table 3). Rate effects on the *NPL* ratio are weak and not robust in sign.

6.4 Sample composition: trims, IE-2015 exclusion, and LOCO influence

We winsorize at the 1–99% tails, exclude IE-2015, and run a leave-one-country-out (LOCO) exercise for the focal elasticities. No single country drives pooled results: LOCO distributions are tight and never flip signs. Trimming and excluding IE-2015 adjust magnitudes slightly but preserve the qualitative message (columns "Trims" and "Drop IE-2015").

6.5 Crisis-window sensitivity: excluding 2008–2009, 2011–2013, and 2020

We re-estimate the baseline FE models excluding the Global Financial Crisis (2008–2009), the sovereign-debt window (2011–2013), and COVID (2020). Directions and economic interpretations persist across outcomes; magnitudes adjust most for NPLs, reflecting information concentration in crisis years.

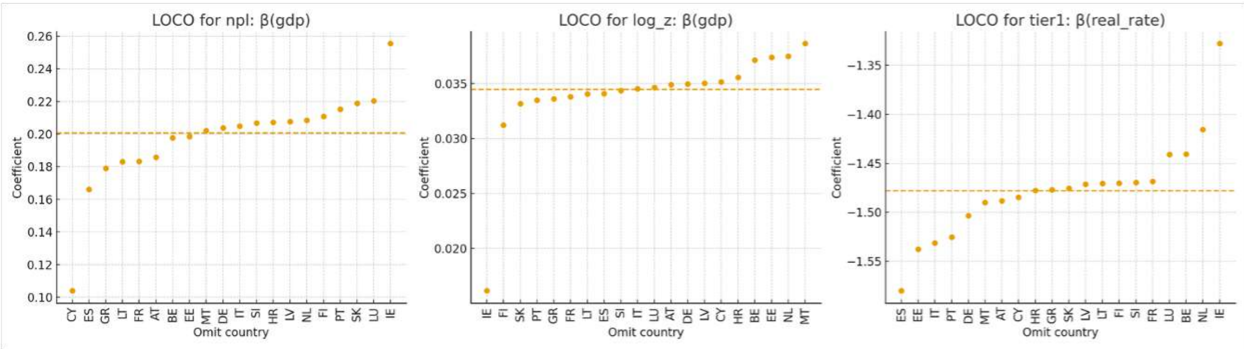

Figure 7: Leave-one-country-out (LOCO) influence on focal elasticities. Points are re-estimated coefficients after omitting the indicated country; dashed line is the baseline.

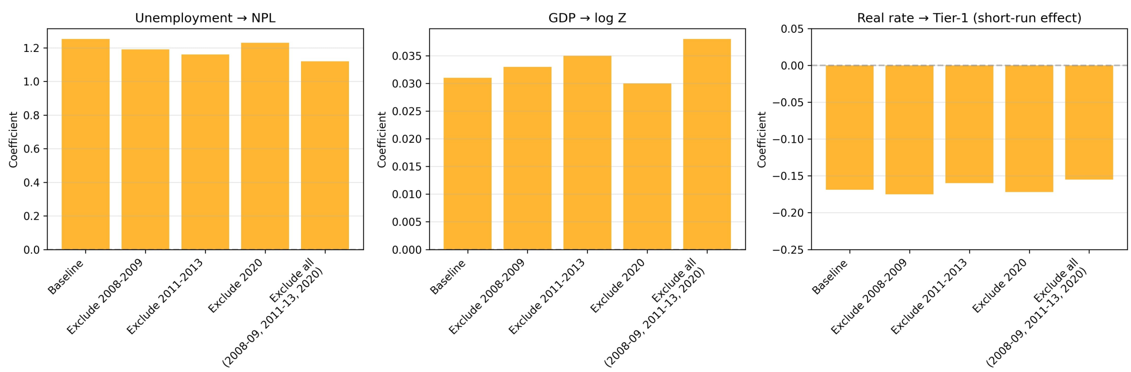

Figure 8: Crisis-window exclusions. Left: Unemployment→NPL (FE). Middle: GDP growth→ log Z (FE). Right: real short rate→Tier-1 *short-run* effect (System–GMM). Bars show point estimates; whiskers denote 95% CIs where available.

6.6 Dynamic identification discipline

We cap instruments (collapsed, shallow lags), report AR(1)/AR(2) and Hansen diagnostics, and enforce #instruments < #countries. Estimates are stable as instruments are compressed (precision falls slightly, as expected). A Difference–GMM variant preserves the sign structure with larger SEs, consistent with fewer valid moments.

6.7 Capital metric robustness: Tier-1 vs. `CAP_TA` (robustness only)

Where available, substituting the unweighted `CAP_TA` ratio mirrors Tier-1 signs (GDP stabilizing; unemployment and rates destabilizing) with different magnitudes due to RWA weighting. We retain Tier-1 as the headline outcome and report `CAP_TA` for completeness.

7 Limitations & Future Work

Our design—a country-level EA-20 panel, 2000–2022—deliberately foregrounds macro variation. This brings clarity but also scope limits.

First, by construction we abstract from within-country bank heterogeneity (business models, funding profiles, liquidity structures). The literature both validates the macro focus and cautions against over-interpreting it: "Loan problems are explained mostly by macroeconomic variables—real GDP growth, unemployment rate, interest rate and public debt—rather than by bank-specific factors" [Chaibi and Ftiti (2015)]. That macro dominance strengthens identification at our level of aggregation but implies that residual micro channels (e.g., liability composition, asset repricing speed) remain in the error term.

Second, cyclical state dependence is strong. Euro-area evidence stresses that "an aggravation in the macroeconomic environment, proxied by sluggish growth and rising unemployment, is closely interrelated with debt-service problems; conversely, improving macroeconomic conditions reduce NPLs" [Anastasiou et al. (2016)]. Likewise, "GDP growth and unemployment emerge as the strongest macroeconomic covariates of European banks' default probabilities" [Soenen and Vennet (2022)]. Our two-way FE and dynamic GMM soak up much confounding co-movement, but in annual data we cannot fully separate contemporaneous shocks to activity, cash flow, and collateral from all balance-sheet responses.

Third, monetary-policy transmission via profitability is only partially captured. Our regressors include a real short rate, but the dataset lacks a consistent term-structure slope and pass-through metrics (e.g., deposit betas, retail-lending-rate convergence). Yet the profitability channel is first-order: "ECB's measures ... have significantly contracted interbank market rates, causing a negative effect on net interest margins ... the extended period of low interest rates and the flattening of the yield curve have significantly eroded the banks' net interest margins" [Angori et al. (2019)]; more generally, "the dominant role of net interest income in bank earnings ... prolonged ... low or even negative interest rates ... may have a negative effect on bank profitability especially for those banks that mainly rely on traditional activity" [Menicucci and Paolucci (2016)]. Because "Monetary policy can affect banks' risk-taking . . . " [Kabundi and Simone (2020)], a richer rate-structure is a priority for future work.

Fourth, Fragmentation and parameter heterogeneity limit pooled interpretations. The literature documents significant heterogeneity across euro-area countries in how macro shocks transmit to banking outcomes. Fragmentation has institutional and market roots—"after 2008, euro-area banks retrenched cross-border claims, reallocating assets from periphery to core countries . . . " with easing after 2012 [Warin and Stojkov (2021)]—and macro-financial co-movements are regime-dependent: "stock and government bond correlations are time-varying and related to growth, inflation, and policy-rate spreads" [Perego and Vermeulen (2016)]. Our interaction work partially addresses this; a fully hierarchical model would do more.

Fifth, credit-boom timing is coarsely measured. We include L1 credit growth and the credit-to-GDP gap, but early-warning performance hinges on multi-year buildups: "Credit expansion in the three years preceding a downturn is the single most important early-warning indicator of bank distress in Europe" [Poghosyan and Čihák (2011)], and "credit growth is a powerful predictor of financial crises" [Bouheni and Hasnaoui (2017)]. Baseline annual lags cannot replicate the full window structure; we leave a systematic lead-length exploration to future work.

Sixth, capital measurement. Tier-1 capital is our primary solvency metric—a risk-weighted ratio standard in euro-area supervision and widely used to gauge capital shortfalls [Jondeau and Sahuc (2022)]. That said, cross-country differences in risk-weighting and simultaneous movements in the numerator (Tier-1 capital via retained earnings) and denominator (risk-weighted assets) can impair comparability. We therefore confine the unweighted CAP_TA measure to robustness checks and flag the need for bank-level supervisory data to disentangle genuine capitalization from RWA re-optimization. Consistent with the reduced-form mechanism we model, deteriorating asset quality erodes earnings and capital: "NPLs . . . reduce banks' profits and, when these are not enough . . . losses consume banks' capital, creating an urgent need for recapitalization" [Anastasiou et al. (2016)].

Seventh, our country panel omits network exposures. Systemic risk also travels via topology: "Complexity, interconnectedness and stability" emphasizes that shock amplification depends on the network itself [Chabot and Bertrand (2021)]. Incorporating cross-border and interbank matrices would refine systemic risk inference beyond NPL/Tier-1 aggregates.

Eighth, institutional and shadow-banking channels are outside the baseline: governance, transparency, regulation, and nonbank finance matter for profitability and resilience [Asteriou et al. (2021)]. Their omission is a conscious simplification, not a claim of irrelevance.

Data constraints. Coverage is high for NPL, Tier-1, and core macro series, but *log Z* ends in 2021 and credit-growth series begin late in several countries (Table 3; Figure 4). Within-country variance is ample (Table 4), and correlations highlight endogeneity risks (Table 5), motivating the dynamic GMM module and our dependence-robust SEs.

Future work. We will (i) bridge to SSM microdata to separate profitability vs. RWA channels for Tier-1; (ii) add term-structure and pass-through measures (building on retail-rate convergence evidence); (iii) embed network exposures using cross-border claims; (iv) integrate institutional and alternative-finance covariates; and (v) estimate event-time responses to multi-year credit booms, consistent with early-warning horizons. These extensions align our reduced-form findings with a more structural account of resilience in the euro-area banking union.

8 Conclusion & Policy Implications

This paper provides a unified, EA-20 baseline mapping from macro conditions to banking stability over 2000–2022 using country–year data, two-way fixed effects, and a dynamic System–GMM layer. Three results organize the evidence.

First, labor-market slack is the primary correlate of realized credit losses. In FE, a 1 pp rise in unemployment is associated with about **+1.25 pp** higher NPLs (p<0.01), with estimates broadly in the **+1.0–+1.3 pp** range across robustness variants (Table 1; Table 7).

Second, the monetary-stance channel transmits into solvency and distance-to-distress. In the dynamic specification (System–GMM), higher real short rates are associated with *lower* log Z (about −0.10 *per +1 pp* in the real short rate; $p\approx0.06$) and *lower* Tier-1 (about −0.17 *per +1 pp*; p>0.10). Given the high persistence of Tier-1, this implies a larger long-run impact.[1] Rate effects on NPLs are *weaker and not robust in sign* (Tables 3–5; Tables 6–8).

Third, the credit cycle leaves an adverse footprint. The **credit-to-GDP gap** robustly predicts subsequent NPL increases in dynamics (Table 4); **lagged credit growth** is adverse for Tier-1 in FE and strengthens at $L2$, while for NPLs its association is *not robust* once persistence is modeled—the gap carries most of the predictive content (Table 7).

These patterns are *stable*. Signs and interpretation survive Driscoll–Kraay errors, regional clustering, alternative lags, trimming, LOCO exclusions (Figure 7), crisis-window drops (Figure 8), and instrument caps/Difference–GMM (Tables 3–5). Together, they support a *macro-dominance* view: within countries, growth, slack, policy rates, and the credit cycle explain a substantial share of movements in stability metrics (cf. Section 7 for scope limits).

Replication. To facilitate further research and policy applications, all data and code are available on GitHub.

[1] Implied long-run effect for Tier-1 $\approx -0.17/(1-0.929) \simeq -2.4$ pp per +1 pp real-rate shock; interpret cautiously given statistical imprecision.

Policy relevance

1. **Countercyclical capital buffer (CCyB).** Positive credit gaps and low slack call for *pre-emptive* activation; slowing activity or rising slack argues for *timely release*. Given Tier-1 persistence, early accumulation of releasable buffers is valuable.
2. **Monitoring dashboard.** Track (i) unemployment, (ii) GDP growth, (iii) short-rate stance (real or nominal with inflation), and (iv) credit-to-GDP gap plus lagged credit growth. Signs are unambiguous; elasticities map scenarios to Z, NPLs, and Tier-1.
3. **Rate-path with prudential complementarity.** Because tighter short rates compress buffers and Z, pair normalization with CCyB space and borrower-based tools, especially when slack is high or credit gaps remain positive.
4. **Borrower-based tools.** Use LTV/DTI to lean against booms and ease them in downturns to avoid amplifying unemployment-driven losses.
5. **Supervisory use.** Apply the elasticities to translate macro scenarios into supervisory metrics for union-wide stress design.

Bottom line. Stronger growth raises distance-to-distress and solvency, rising unemployment lifts NPLs, tighter policy rates compress buffers and Z, and credit booms presage weaker stability. Because these mappings survive extensive robustness and dynamic checks, they can be read into CCyB settings, borrower-based policies, and the rate path—providing a compact, evidence-based toolkit for macro-financial stabilization.

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